

ARATRIK BOSE

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with nearly 4 years of production experience building **Generative AI** systems — including LLM-powered conversational agents, retrieval-augmented generation (RAG) pipelines, and real-time voice and avatar systems for safety-critical automotive environments. Combines applied research — including 2 peer-reviewed publications, optimizing inference pipelines for edge deployment, and hardening production codebases against security risk. Skilled at translating complex AI concepts for cross-functional and non-technical stakeholders.

PROFESSIONAL EXPERIENCE

AI Engineer — Tata Consultancy Services, AI Mobility CoE

2024 – Present

- Reverse-engineered an undocumented 21-dimension latent expression space inside a third-party talking-head model — mapping individual keypoint indices to specific **facial expressions** and identifying a masking bug that was silently discarding emotion-conditioned outputs before rendering — enabling an **emotion-aware avatar** that now dynamically responds with three distinct expressions based on user input.
- Designed and optimized AI voice agents — including prompt engineering, dialogue-flow design (intent parsing, state management, status-driven response logic), and specialized speech, emotion, and accent recognition models — achieving **730–800 ms** end-to-end inference latency, including deployment to NVIDIA Jetson Orin edge hardware via **FP16/INT8 TFLite quantization**.
- Diagnosed a critical TTS pipeline bottleneck, isolating model-reload overhead as **~88% of total per-request latency** (2.9s of 3.3s), and re-architected the backend around persistent subprocesses (thread-safe locking, auto-restart, file-diffing) and RESTful/FastAPI + IPC orchestration — eliminating reload cost from the hot path while powering real-time voice control, telemetry, and avatar-driven interaction features.
- Worked on research and development for AI-driven automotive and intelligent systems, applying deep learning and image processing architectures across 3+ core in-cabin monitoring and driver assistance projects, cutting overall processing time from **115s to 30s (a ~74% reduction)**.
- Designed and deployed Generative AI-powered conversational assistant that runs locally behind a Python/FastAPI backend, with a retrieval-augmented generation (RAG) layer that scores and injects live web/news context into responses for improved relevance and reduced hallucination even when **running offline**.
- Audited and secured the codebase architecture of a patented, on-road deployed tool used by multiple customers, identifying and mitigating **200+ potential security** vulnerabilities and helping upgrade customer software packages to **ISO 26262** compliance.

Technical Developer — Tata Consultancy Services

2022 – 2024

- Led a cross-functional engineering team of 5 developers to design and deliver an end-to-end manufacturing execution systems management tool (SQL Server, HTML/CSS) for a Japan-based printer manufacturing company.
- Streamlined factory floor execution workflows using Delmia Apriso, improving operational efficiency and shop-floor data visibility.

TECHNICAL SKILLS

AI/ML: Generative AI, Deep Learning, Machine Learning, LLMs, Speech Recognition, Voice AI, TTS, Computer Vision, GAN, RAG

Frameworks: PyTorch, TensorFlow, OpenCV, Hugging Face, Whisper, Wav2Vec2, ONNX

Languages: Python, Embedded C, C++, Java, MATLAB, SQL, HTML, CSS

Edge/Deployment: NVIDIA Jetson Orin, Jetson Thor, TFLite (FP16/INT8 Quantization)

Tools: Git, JIRA, SQL Server, Delmia Apriso

PUBLICATIONS

- EEG-Based Brain-Computer Interface for Prosthetic Limbs — Taylor & Francis. DOI: 10.1080/10255842.2025.2602830
Proposed a signal-processing-based magnification to amplify subtle EEG patterns, combined with machine-learning classifiers — reaching up to 98% accuracy, aimed at improving accessibility for amputees.
- Malicious Transaction URL Detection Using Logistic Regression — Springer. DOI: 10.1007/978-981-99-4433-0_8
Built a logistic-regression-based classifier to detect malicious URLs in financial transactions, improving fraud-detection accuracy to around 99%.

EDUCATION

M.Tech. in Automotive Electronics, Amrita Vishwa Vidyapeetham

2024 – 2026

B.Tech. in Computer Science & Engineering, Dr. B.C Roy Engineering College

2018 – 2022